DNA/Protein Sequence Analysis using BLAST

Introduction

Bioinformatics is an interdisciplinary field that combines biology and computational tools to analyze biological data. One of the most widely used bioinformatics tools is the Basic Local Alignment Search Tool (BLAST), developed by the National Center for Biotechnology Information (NCBI). BLAST enables researchers to compare DNA or protein sequences with existing database sequences, identify homologous proteins, and study evolutionary relationships.
In this analysis, the Human Insulin protein sequence was retrieved from the UniProt database and analyzed using the NCBI Protein BLAST (BLASTP) tool to identify similar protein sequences and evaluate sequence conservation among different organisms.

Objective

The objective of this study is to analyze the Human Insulin protein sequence using the NCBI Protein BLAST tool and identify homologous protein sequences by comparing them with sequences available in the NCBI protein database.

Materials and Software Used

- *UniProt Protein Database*
- *NCBI Protein BLAST (BLASTP)*
- *Internet Browser*

Methodology

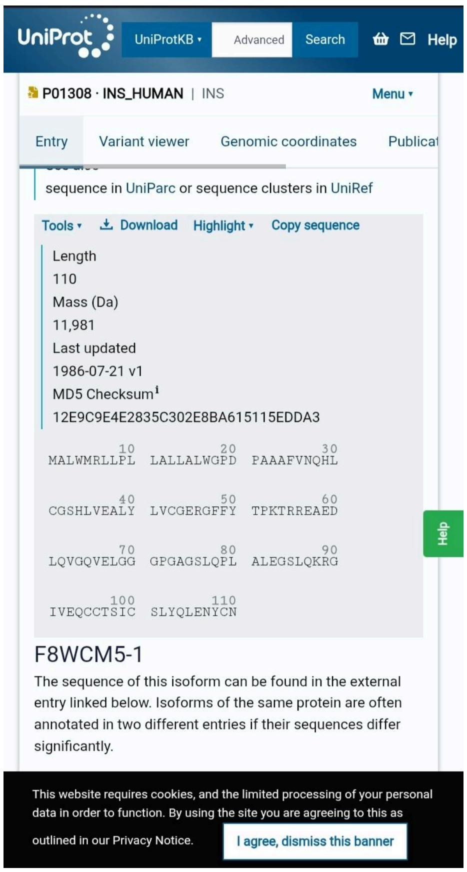
The Human Insulin protein sequence was obtained from the UniProt database. The retrieved protein sequence was copied and submitted to the NCBI Protein BLAST (BLASTP) program using the default search parameters. The BLAST search compared the query sequence with protein sequences available in the NCBI database. The resulting alignments were examined based on query coverage, percentage identity, bit score, and E-value to determine the most significant homologous protein sequence.

Results and Discussion

The BLAST analysis successfully identified multiple homologous insulin protein sequences from different organisms. The highest-scoring match corresponded to the insulin preproprotein of Pan troglodytes (Chimpanzee). The alignment showed 98% sequence identity and 99% positive matches, with an E-value of 1e-72 and a Bit Score of 221, indicating an extremely significant sequence similarity. These results demonstrate that the insulin protein is highly conserved among closely related species and emphasize the importance of BLAST in comparative sequence analysis and evolutionary studies.

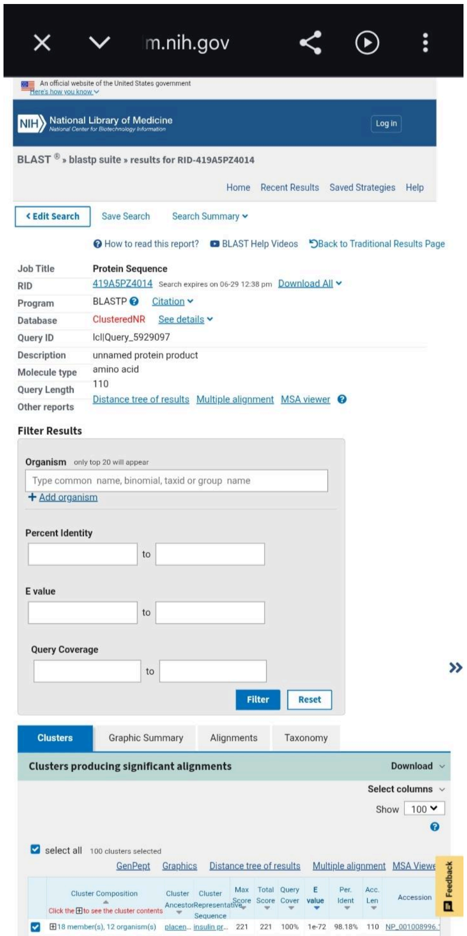
Screenshots

☐ *UniProt Protein Sequence*



Human Insulin protein sequence retrieved from the UniProt database

☐ *BLAST Result Summary*



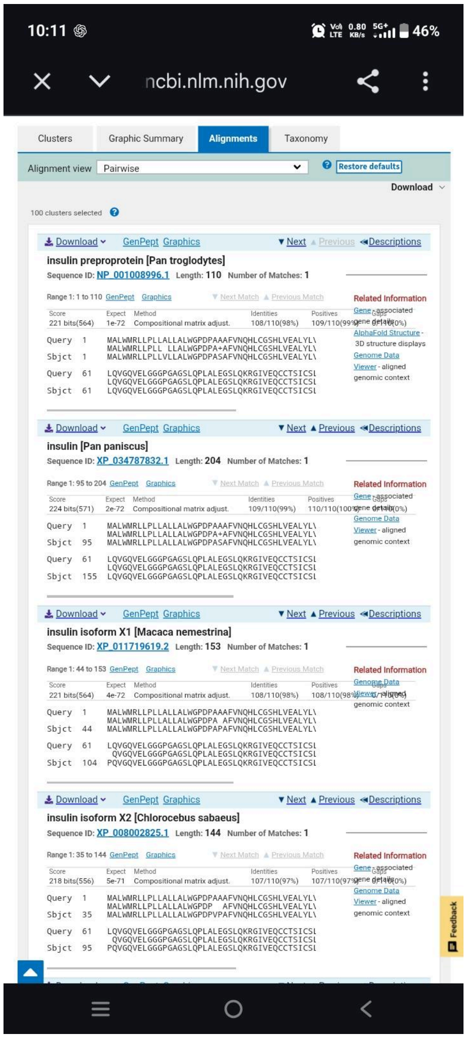
BLAST search summary of the Human Insulin protein sequence.

BLAST Result Table

Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
18 member(s), 12 organism(s)	placen...	insulin pr...	221	221	100%	1e-72	98.18%	110	NP_001008996.1
3 member(s), 3 organism(s)	primat...	insulin [P...	224	224	100%	2e-72	99.09%	204	XP_034787832.1
5 member(s), 5 organism(s)	primat...	insulin is...	221	221	100%	4e-72	98.18%	153	XP_011719619.2
1 member(s), 1 organism(s)	green ...	insulin is...	218	218	100%	5e-71	97.27%	144	XP_008002825.1
1 member(s), 1 organism(s)	black s...	insulin is...	218	218	100%	1e-70	97.27%	147	XP_017743290.1
1 member(s), 1 organism(s)	green ...	insulin is...	218	218	100%	7e-70	97.27%	209	XP_008002752.1
5 member(s), 5 organism(s)	primat...	INS isofo...	191	191	100%	1e-60	88.18%	98	PNJ16423.1
2 member(s), 2 organism(s)	placen...	insulin [T...	184	184	100%	4e-58	90.00%	110	KAN7991649.1
1 member(s), 1 organism(s)	Europe...	insulin [M...	184	184	100%	5e-58	79.09%	110	XP_080768357.1
1 member(s), 1 organism(s)	other s...	ZZ-proins...	183	183	78%	3e-56	100.00%	210	UFQ90408.1
4 member(s), 4 organism(s)	bats	insulin [R...	178	178	100%	2e-55	88.18%	110	KAN3685427.1
1 member(s), 1 organism(s)	high G...	hypotheti...	179	179	78%	2e-55	100.00%	143	MCZ4612290.1
2 member(s), 1 organism(s)	other s...	HPPI [syn...	175	175	78%	1e-54	98.84%	97	AGG11869.1
member(s), organism(s)	Rhesu...	insulin,pr...	174	174	78%	3e-54	98.84%	86	1006230A
1 member(s), 1 organism(s)	reed vo...	insulin [Al...	175	175	100%	5e-54	85.45%	128	XP_049992438.1
1 member(s), 1 organism(s)	Alston'...	hypotheti...	175	175	100%	9e-54	85.45%	145	KAL6031003.1
1 member(s), 1 organism(s)	other s...	SUMO-Li...	176	176	78%	1e-53	97.67%	207	UIK24472.1
3 member(s), 2 organism(s)	rabbits...	insulin pr...	173	173	100%	2e-53	86.36%	110	NP_001353985.1
1 member(s), 1 organism(s)	copper...	Insulin [Pl...	175	175	100%	2e-53	87.27%	184	KAL0628984.1
1 member(s), 1 organism(s)	E. coli	hypotheti...	172	172	95%	3e-53	84.76%	109	HDJ0545968.1
1 member(s), 1 organism(s)	Chines...	Insulin [C...	177	177	100%	3e-53	86.36%	248	EGW08477.1
1 member(s), 1 organism(s)	Bank v...	hypotheti...	172	172	100%	6e-53	84.55%	135	KAK7801657.1
2 member(s), 2 organism(s)	New W...	insulin is...	173	173	100%	6e-53	86.36%	155	XP_064226392.1
2 member(s), 2 organism(s)	primat...	insulin is...	171	171	100%	8e-53	84.55%	110	XP_003798420.1
4 member(s), 4 organism(s)	New W...	RecNam...	171	171	100%	9e-53	85.45%	108	P67972.1
8 member(s), 7 organism(s)	root	insulin-1 ...	171	171	100%	1e-52	82.73%	110	NP_062002.1
1 member(s), 1 organism(s)	other s...	NanoLuc-...	177	177	100%	2e-52	81.82%	332	XOB88773.1
1 member(s), 1 organism(s)	black fl...	insulin [P...	170	170	100%	3e-52	84.55%	110	XP_006911039.1
10 member(s), 10 organism(s)	placen...	insulin pr...	170	170	100%	3e-52	88.18%	110	NP_001123565.1
36 member(s), 29 organism(s)	root	insulin-2 ...	169	169	100%	4e-52	81.82%	110	NP_001172012.1
4 member(s), 4 organism(s)	primat...	insulin [E...	169	169	100%	4e-52	87.27%	109	XP_069327944.1
1 member(s), 1 organism(s)	red fox	insulin is...	169	169	100%	6e-52	88.18%	124	XP_072614830.1
1 member(s), 1 organism(s)	North ...	insulin [Er...	168	168	100%	2e-51	80.91%	110	KAM6160390.1
1 member(s), 1 organism(s)	white-t...	insulin is...	169	169	100%	2e-51	85.45%	148	XP_035119627.2
1 member(s), 1 organism(s)	striped...	insulin [H...	167	167	100%	4e-51	83.64%	110	XP_039081000.1
1 member(s), 1 organism(s)	grey w...	Insulin [E...	169	169	100%	5e-51	88.18%	177	MBW01439.1
1 member(s), 1 organism(s)	other s...	proinsuli...	166	166	77%	8e-51	94.38%	105	AWO14628.1
4 member(s), 4 organism(s)	rodents	insulin-1-l...	166	166	100%	1e-50	78.18%	110	XP_052028151.1
1 member(s), 1 organism(s)	Iriomot...	insulin is...	167	167	100%	2e-50	81.82%	171	GAB5578225.1
3 member(s), 2 organism(s)	gerbils	RecNam...	165	165	100%	2e-50	80.91%	110	Q62587.1
1 member(s), 1 organism(s)	North ...	LOW QUA...	169	169	100%	2e-50	88.18%	234	XP_061059793.1
5 member(s), 4 organism(s)	Old Wo...	insulin is...	164	164	100%	9e-50	85.45%	135	XP_025211012.1
6 member(s), 6 organism(s)	cat fa...	insulin pr...	163	163	100%	1e-49	80.91%	110	NP_001009272.1
1 member(s), 1 organism(s)	gray se...	insulin [H...	164	164	100%	2e-49	85.45%	148	XP_035930639.2
1 member(s), 1 organism(s)	greater...	insulin [R...	162	162	100%	4e-49	89.09%	123	XP_032975469
1 member(s), 1 organism(s)	meerkat	insulin [S...	161	161	100%	8e-49	86.36%	110	XP_029770545
9 member(s), 9 organism(s)	rodents	insulin pr...	161	161	100%	9e-49	89.09%	110	NP_001269184
1 member(s), 1 organism(s)	naked ...	insulin [H...	160	160	100%	1e-48	79.09%	110	XP_004852149
1 member(s), 1 organism(s)	other s...	SEAP-P2...	173	173	100%	2e-48	81.82%	652	WVR21740.1

Homologous protein sequences identified by NCBI BLAST.

BLAST Alignment



Pairwise sequence alignment of Human Insulin with the top homologous protein

Conclusion

The Human Insulin protein sequence was successfully analyzed using the NCBI Protein BLAST tool. The analysis identified highly homologous protein sequences with excellent alignment statistics, confirming the evolutionary conservation of the insulin protein. This study demonstrates the effectiveness of BLAST as a reliable tool for sequence similarity analysis, protein identification, and comparative bioinformatics research.

References

1.UniProt Consortium. UniProt Knowledgebase (UniProtKB). *https://www.uniprot.org/�*

2.National Center for Biotechnology Information (NCBI). Basic Local Alignment Search Tool (BLAST).

https://blast.ncbi.nlm.nih.gov/�